

LESSON 4-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Percent Means Per Hundred**

Maryland Standard 6.RP.A.3c • Learning target – I can write a ratio as a percent by finding the equivalent ratio out of 100.

REMEMBER

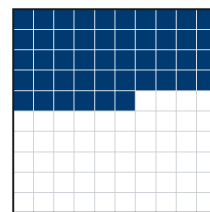
- **Percent** means “per hundred.” $37\% = 37$ out of 100.
- On a hundred grid, each square is **1%**.
- Make an **equivalent ratio** out of 100: multiply **both** terms by the same number.

WORKED EXAMPLE

Out of every 20 students, 7 ride the bus. What percent of the students ride the bus?

Step 1 $20 \times 5 = 100$, so multiply both terms by 5.**Step 2** $7 \times 5 = 35$, so 7 out of 20 = 35 out of 100.**Step 3** 35 out of 100 = 35%.**Answer** **35%****WORD BANK****percent** – out of 100**%** – the percent sign**whole** – all of it; the whole is 100%**equivalent ratio** – the same comparison with new numbers**PRACTICE***show your work • label units*

1. The grid has 100 equal squares.

Step 1 Count the shaded squares. Each square is 1%.**Step 2** Subtract from 100% to find the part not shaded.**a.** What percent of the grid is shaded? _____ %**b.** What percent is **not** shaded? _____ %

2. Write each ratio as a percent. First scale it to an equivalent ratio out of 100.

Step 1 Ask: what times the second number makes 100?**Step 2** Multiply the first number by that same number.**a.** 3 out of 10 = _____ % **b.** 7 out of 25 = _____ % **c.** 1 out of 2 = _____ %**d.** 17 out of 50 = _____ % **e.** 2 out of 5 = _____ % **f.** 3 out of 20 = _____ %**g.** 13 out of 25 = _____ % **h.** 9 out of 10 = _____ %

- 3.
- Animal Shelter.**
- Out of every 25 dogs at a shelter, 9 are puppies. Complete the ratio table. Then write the percent of dogs that are puppies.

Puppies	9	18		
All dogs	25	50	75	100

Percent of puppies _____ %

4. Garden Plants. A garden has **50 plants**. **22** of them are tomato plants.

Step 1 $50 \times 2 = 100$. Multiply 22 by 2 too.

Step 2 Subtract your percent from 100%.

a. What percent are tomato plants? _____ %

b. What percent are **not** tomato plants? _____ %

5. Find the Error. To write 7 out of 20 as a percent, Ana added 80 to both numbers. She got 87 out of 100, so she wrote 87%. Explain her mistake and find the correct percent.

Frame Ana's mistake is _____. $20 \times \underline{\hspace{1cm}} = 100$, so $7 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$, which is ____%.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Find a percent on a label, sign, or screen at home. Tell someone what it means "out of 100."

LESSON 4-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Percents Greater Than 100%**

Maryland Standard 6.RP.A.3c • Learning target – I can use percents greater than 100% to describe an amount larger than the whole.

REMEMBER

- **100%** is one whole. Over 100% is **more than the whole**.
- $200\% = 2$ times the whole. $150\% = 1.5$ times.
- “Times as much” to percent: multiply by 100. 2.5 times = 250%.

WORKED EXAMPLE

A puppy grew from 10 lb to 13 lb. Write 13 lb as a percent of 10 lb (the whole).

Step 1 Scale by 10: 13 to 10 = 130 to 100.

Step 2 130 out of 100 = 130%, more than the whole.

Answer 130%

WORD BANK

whole – the amount that is 100%

more than 100% – more than the whole

times as much – 2 times = 200%, 3 times = 300%

PRACTICE

show your work • label units

1. Write each amount as a percent of the whole.

Step 1 1 time as much is 100%.

Step 2 Multiply the number of times by 100%.

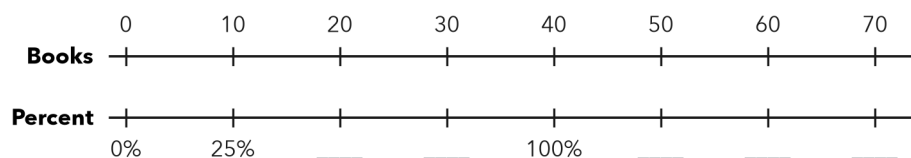
a. 3 times as much = ____ %

b. 1.75 times as much = ____ %

2. **Library Books.** Monday (40 books) is the whole. Tuesday: 70 books. Fill in the double number line. Write Tuesday as a percent of Monday.

Step 1 40 books is 100%, so 10 books is 25%.

Step 2 Count up by 25% for each 10 books.



Tuesday _____ %

3. **Bake Sale.** A class set a goal of \$200. They raised \$260. What percent of the goal did they raise?

Step 1 \$200 is the whole, 100%.

Step 2 Divide both by 2: 260 to 200 is the same as ____ to 100.

Answer _____ %

4. **Chess Club.** Last year a chess club had 20 members (the whole). This year it has 33 members. Write this year's members as a percent of last year's.

Answer _____ %

- 5. Find the Error.** Malik says 60 cm is 60% of 30 cm. Explain his mistake and fix it.

Frame Malik is wrong because 60 is ___ times 30, so 60 cm is ___% of 30 cm.

TEST PRACTICE

choose the best answer

- 6.** 45 points is what percent of 30 points?

A 15%

B 45%

C 115%

D 150%

- 7.** 250% of an amount is how many times the amount?

A 0.25 times

B 2.5 times

C 25 times

D 250 times

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #_____
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AT HOME Tell someone at home what 200% of their age would be.